

AI-Semiconductor Workshop Evaluation Project

Project Objectives

The purpose of this project is to evaluate the effectiveness of teaching techniques used in a technical workshop by examining participant experiences, learning processes, and outcomes. This evaluation is designed to inform improvements to future workshop offerings within this specific context.

The objectives of this project are:

1. To characterize the diversity of participants in terms of background, prior experience, and learning goals.
2. To examine how different types of adult learners experience the workshop, including their preferences for instructional methods such as guided instruction, hands-on activities, and independent work.
3. To evaluate the effectiveness of teaching techniques in supporting learning, engagement, and understanding of technical material.
4. To identify barriers and challenges that participants encounter during the workshop, including issues related to pace, prior knowledge, and tools.
5. To assess changes in participant confidence and perceived ability to learn independently over the course of the workshop.
6. To determine how well the workshop aligns with participants' personal, educational, and career goals.
7. To generate actionable insights and recommendations for improving the design, structure, and delivery of future workshops.

Evaluation Plan

Design

This project uses a program evaluation design to assess participant perceptions of teaching techniques in a 5-day technical workshop. The evaluation focuses on understanding how instructional approaches support learning, engagement, and the development of independent problem-solving skills among adult learners.

This activity is conducted for the purpose of improving a specific educational program and is not intended to contribute to generalizable knowledge beyond this context.

Participants

Participants include adult learners enrolled in the workshop, with diverse backgrounds such as occupational/technical programs and pre-engineering pathways. Participation in the evaluation is voluntary and anonymous.

Data Collection

Data will be collected using two anonymous surveys:

- **Pre-Workshop Survey (Day 1):**
 - Participant background and prior experience
 - Learning goals and expectations
 - Preferred learning methods
 - Anticipated challenges and baseline confidence
- **Post-Workshop Survey (Day 5):**
 - Perceived effectiveness of teaching techniques
 - Evaluation of learning methods (e.g., hands-on, guided, independent)
 - Engagement and participation
 - Experienced barriers and challenges
 - Changes in confidence and independent learning ability
 - Overall program value and recommendations

Survey responses will include a combination of multiple-choice, Likert-scale, and open-ended questions.

Key Variables

The evaluation focuses on the following variables:

- **Participant Characteristics:** background, prior experience, goals
- **Teaching Features:** clarity, structure, hands-on learning, guidance vs independence
- **Learning Experience:** engagement, relevance, participation
- **Barriers:** pace, prior knowledge, tools, time constraints
- **Outcomes:** confidence, independence, perceived learning
- **Program Value:** satisfaction, usefulness, future intentions

Data Analysis

Quantitative Analysis:

- Descriptive statistics (frequencies, percentages, averages)
- Comparison of pre- and post-workshop responses (e.g., confidence levels)
- Group comparisons based on participant background (e.g., occupational vs pre-engineering)

Qualitative Analysis:

- Thematic analysis of open-ended responses
- Identification of recurring themes such as:
 - effectiveness of hands-on learning
 - need for guidance vs independence
 - common barriers
 - suggestions for improvement

Ethical Considerations

- Participation is voluntary
- No identifying information is collected
- Data is anonymous
- Results are used solely for program evaluation and improvement

Expected Use of Results

Findings from this evaluation will be used to:

- Improve the balance between guided instruction and independent learning
- Adjust workshop pace and structure
- Enhance hands-on and experiential learning components
- Better support participants with diverse backgrounds and needs

Pre-Workshop Survey

Instructions: This survey is anonymous and will be used only to evaluate and improve the workshop. Please answer honestly. Please enter the identification number provided to you. This will be used to match your pre- and post-survey responses. Your responses will remain anonymous.

1. Participant ID: -----

Math and Concept Background

1. Which of the following concepts are you familiar with? (Select all that apply)

- Solving algebraic equations
- Interpreting graphs (e.g., slope, trends)
- Linear relationships (e.g., $y = mx + b$)
- Systems of equations
- Vectors (magnitude, direction)
- Matrices (basic operations)
- Basic statistics (mean, variance)
- None of the above

2. How confident are you in using the following concepts?

- Graphs and trends:
Not confident (1) (2) (3) (4) (5) Confident
- Algebra (equations):
Not confident (1) (2) (3) (4) (5) Confident
- Linear relationships:
Not confident (1) (2) (3) (4) (5) Confident
- Vectors:
Not confident (1) (2) (3) (4) (5) Confident
- Matrices:
Not confident (1) (2) (3) (4) (5) Confident
- Statistics:
Not confident (1) (2) (3) (4) (5) Confident

3. How prepared do you feel for learning data analysis or machine learning concepts?

- Not prepared
- Slightly prepared
- Moderately prepared
- Well prepared

1. Learner Background

1. What is your current program?

- Occupational / Technical Program
- Pre-engineering Program
- Other: -----

2. How much prior experience do you have with this topic?

- None
- Beginner
- Intermediate
- Advanced

3. How often do you use technical or problem-solving skills in real life or work?

- Never(1) Rarely(2) Sometimes(3) Often(4) Very Often(5)

2. Goals and Expectations

1. What are your main goals for attending this workshop? (Select all that apply)

- Gain practical skills
- Prepare for further education
- Explore interest in the topic
- Build confidence
- Other: _____

2. I expect this workshop to be relevant to my personal or professional goals.

- Strongly Disagree (1) Disagree (2) Neutral (3) Agree (4) Strongly Agree (5)

3. How soon do you expect to use what you learn?

- Immediately
- Soon
- Not sure

3. Learning Preferences

1. Which methods help you learn best? (Select all that apply)

- Lecture
- Demonstration
- Hands-on practice
- Group work

2. How much guidance do you prefer when learning something new?

- Very guided (1) (2) (3) (4) Very independent (5)

4. Barriers and Readiness

1. What challenges might affect your learning? (Select all that apply)

- Time constraints
- Lack of prior knowledge
- Confidence
- Technology/tools
- None

2. I feel confident learning new technical material independently.

- Strongly Disagree (1) Disagree (2) Neutral (3) Agree (4) Strongly Agree (5)

3. I feel confident solving technical problems on my own.

- Strongly Disagree (1) Disagree (2) Neutral (3) Agree (4) Strongly Agree (5)

5. Open-Ended Questions

1. What do you hope to gain most from this workshop?

2. What concerns (if any) do you have before starting?

Post-Workshop Survey

Instructions: This survey is anonymous and will be used to evaluate and improve the workshop. Please enter the same identification number you used in the pre-survey.

1. Participant ID: _____

Concept Understanding and Learning Outcomes

1. How would you rate your understanding of the following concepts after the workshop?

- Linear relationships (regression):
Not at all (1) (2) (3) (4) Strong (5)
- Working with data (patterns, trends):
Not at all (1) (2) (3) (4) Strong (5)
- Dimensionality reduction (PCA idea):
Not at all (1) (2) (3) (4) Strong (5)
- Matrix-based thinking:
Not at all (1) (2) (3) (4) Strong (5)

2. Compared to before the workshop, your understanding of math concepts used in this workshop is:

- Much lower
- Slightly lower
- About the same
- Slightly higher
- Much higher

3. Which concepts were most difficult for you?

4. How well did your prior math knowledge prepare you for this workshop?

- Not at all
- Slightly
- Moderately
- Well
- Very well

5. How much time did you spend actively working on workshop tasks?

- Very little
- Some
- Moderate
- A lot

Open-Ended Reflection

1. What prior knowledge helped you the most during this workshop?
2. What background knowledge do you wish you had before attending?

1. Teaching and Learning Experience

1. Which aspects of the workshop helped your learning the most? (Select up to 3)
 - Instructor explanations
 - Demonstrations
 - Hands-on activities
 - Working with real data
 - Group collaboration
 - Independent practice
2. Which aspect was **least helpful** for your learning?
 - Instructor explanations
 - Demonstrations
 - Hands-on activities
 - Group collaboration
 - Independent work
 - None (all were helpful)

3. How would you describe the overall structure of the workshop?

- Very well structured
- Mostly clear and organized
- Somewhat unclear at times
- Difficult to follow

4. How would you describe the pace of the workshop?

- Too slow
- Slightly slow
- Appropriate
- Slightly fast
- Too fast

2. Learning Approach and Fit

1. What type of learning approach worked best for you?

- Mostly guided (step-by-step instruction)
- A mix of guidance and independent work
- Mostly independent exploration

2. Did you feel you had the right balance between guidance and independence?

- Yes, it was well balanced
- I needed more guidance
- I needed more independence

3. How relevant was the workshop to your personal or career goals?

- Very relevant
- Somewhat relevant
- Slightly relevant
- Not relevant

4. How applicable was the content to real-world situations?

- Very applicable
- Somewhat applicable
- Slightly applicable
- Not applicable

3. Challenges and Barriers

1. What challenges did you experience? (Select all that apply)

- Pace was too fast
- Lack of prior knowledge
- Difficulty with tools or technology
- Instructions were unclear
- Time or fatigue

- No major challenges
- Other (Please explain)

—

2. How would you rate the difficulty level?

- Too easy
- Slightly easy
- Appropriate
- Slightly difficult
- Too difficult

4. Confidence and Learning Growth

1. Compared to before the workshop, your confidence in learning technical material is:

- Much lower
- Slightly lower
- About the same
- Slightly higher
- Much higher

2. Your ability to solve problems independently is now:

- Much lower
- Slightly lower
- About the same
- Slightly higher
- Much higher

5. Engagement and Participation

1. How actively did you participate in the workshop?

- Very actively
- Moderately actively
- Occasionally
- Rarely

2. Which activities engaged you the most? (Select up to 2)

- Hands-on work
- Group collaboration
- Instructor explanation
- Independent tasks

6. Overall Program Evaluation

1. Overall, how would you rate this workshop?
 - Excellent
 - Good
 - Fair
 - Poor
2. Would you recommend this workshop to others?
 - Definitely
 - Probably
 - Not sure
 - Probably not
 - Definitely not
3. What are you most likely to do next?
 - Apply these skills in a job
 - Take more technical courses
 - Pursue an engineering degree
 - Continue exploring

7. Open-Ended Questions

1. What helped your learning the most? Why?
2. What would you improve about the workshop?
3. How could this workshop better support learners like you?